CoRE Stack · Tower Services

STACD

Provenance for geospatial compute: every output records the algorithm and parameters that made it.

Saharsh Laud · Supervisor: Prof. Aaditeshwar Seth · IIT Delhi · October 2026

Services produce data, but not its history

Many services

Drone crowns, bioacoustics, LULC and forecasts each run their own compute.

No common record

Outputs sit in folders with no link to the code or settings behind them.

Hard to reproduce

Test runs and final runs look alike, so results are hard to trust or repeat.

STACD = STAC + DAG lineage

STAC

A standard catalog of the datasets: map, assets and metadata for each output.

STAC-D

A catalog of the workflows: DAGs, algorithms, and which datasets they read and write.

Each dataset links to the algorithm version that produced it, and each algorithm links to its outputs.

How a run flows through the platform

1

Service backend

Exposes one compute API.

2

Airflow DAG

Generated from YAML; calls the API.

3

STACD register

Versions the output in the database.

4

STAC item

Written to the catalog with provenance.

5

Browsers

STAC and STACD browsers show data and lineage.

All compute goes through Airflow and STACD. Onboarding a service needs three YAML files: DAG, algorithm repo and dataset repo.

What gets recorded for every output

Field	Why it matters
Algorithm and version	Which method produced the data
DAG and run id	Which workflow run, and when
Input parameters	Region, year, settings: enough to repeat the run
Dataset version	Reruns create new versions instead of overwriting
Asset id	Where the output itself lives

Services currently running on STACD

Service	Output	Catalog layout
Drone Imagery Analysis	Tree crowns and species maps	By project
Bioacoustic Data	Bird detections and indices	Project, algorithm, job
Do It Yourself LULC	10 m land-cover rasters	Custom LULC items
Pest forecast, layer maps	Forecasts and CoRE Stack layers	Per pipeline

More services to come soon...

Reproducible, composable, traceable

Public outputs are browsable in the STAC browser. One click leads to the STACD algorithm, its version and the recorded parameters.

STACD paper: Link to paper

CoRE Stack: core-stack.org

Landscape analytics: explorer.core-stack.org